

● *Original Contribution***ACOUSTIC DROPLET VAPORIZATION FOR THERAPEUTIC AND
DIAGNOSTIC APPLICATIONS**OLIVER D. KRIPFGANS, J. BRIAN FOWLKES, DOUGLAS L. MILLER, O. PETTER ELDEVIK and
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Abstract—A phase shift droplet emulsion is introduced as an aid to unusual ultrasound (US) applications. The transpulmonary droplet emulsion (90% < 6 μm diameter) is made by mixing saline, bovine albumin and dodecafluoropentane. It has been observed that an acoustic pressure threshold exists, above which the droplets vaporize into bubbles approximately 25 times the original diameter. For frequencies between 1.5 and 8 MHz, the threshold decreases from 4.5 to 0.75 MPa peak rarefactional pressure. This paper presents preliminary results for droplet preparation and their evaporation as a function of applied acoustic pressure and frequency, as well as simulations of the lifetime of these gas bubbles based on gas diffusion. *In vivo* experiments were simulated by the evaporation of droplets in blood flowing under attenuating material. We propose that this agent might be useful for tissue occlusion in cancer treatment, as well as for phase aberration corrections in acoustic imaging.
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Key Words: Contrast agent, Phase shift agent, Cancer treatment, Phase aberration, Medical imaging.

INTRODUCTION AND LITERATURE

Boehm et al. (1997) showed that it is possible to “starve” cancer cells to death by restricting their blood supply. Angiostatin and endostatin suppress the development of supply arteries that tumors attract chemically and need for their own growth. Occlusion therapy has also been used to control blood flow, including direct injection of material into arteries, which either blocks blood flow at the injection site or immediately downstream. A high density of large solid emboli (40–50 μm) are clearly occlusive and produce permanent damage to tissue (Lacourreye et al 1993). Many different materials have been introduced by intravascular catheters and percutaneously, to occlude flow. These materials have been either absorbable, including blood clot, cellulose and gelatin sponge (Gelfoam), or nonabsorbable including particulates, fluids, coils, balloons and streamers. Hepatocellular carcinoma, occurring in 2–30 per 100,000 males each year, has been treated successfully with embolotherapy where localized drug delivery was included in the procedure (Nakagawa and Castaneda-Zuniga 1997;

Di Segni et al. 1997). Renal carcinoma is another condition amenable to embolotherapy. It is the most common cancer in the urinary tract and accounts for ~ 3% of malignancies overall. Several reports have indicated successful treatments by embolization including 78% tumor necrosis (Sandoval et al. 1983). One of the more important factors in the success of such therapies is the ability to direct and confine the treatment of a targeted region. The techniques described here offer a potentially convenient method to target multiple vessels or a capillary bed for occlusion following a simple IV administration of the perfluorocarbon droplets (specifically dodecafluoropentane, DDFP). When acoustic droplet vaporization (ADV) is desired in feeder vessels, target vessels could be identified by a method (currently under development in our laboratory) known as contrast interruption (Fowlkes et al. 1998), where the flow of conventional ultrasound (US) contrast agent can be halted in specific arteries by the targeted application of low amplitude US.

Laplace pressure compresses bubbles made from droplets and increases the diffusion of the internal gas (mainly gaseous DDFP) into the host liquid. However, due to the low solubility and diffusibility of perfluorocarbon gases in water (Table 1), bubbles filled with them

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Table 1. Constants used in simulation of bubble growth and dissolution.

Gas	Diffusion coefficient $D \cdot 10^5 \text{ cm}^2 \text{ s}^{-1}$	Water solubility $C_{\infty} \cdot 10^{-7} \text{ mole mL}^{-1}$	% gas content in saline
N ₂	2.00	5.84	76.1
CO ₂	1.91	267.5	0.697
O ₂	2.42	11.04	22.3
Ar	2.5	12.09	0.934
C ₅ F ₁₂	0.6236*	0.0401 [†]	0

Diffusion and solubility constants were taken from Lide (2000), except those for C₅F₁₂ (see footnote).

* Computed using molar volume of 177.8 mL mole⁻¹ (Simons and Dunlap 1950); [†] taken from Kabalnov et al (1998, 1998b).

have considerably longer lifetimes in water (and blood) compared to air bubbles of the same size. Therefore, large numbers of relatively large perfluorocarbon gas bubbles should provide extended occlusion, potentially sufficient to infarct tissue (Lipton 1999). Contrast agents containing small perfluorocarbon gas bubbles are already in use (*i.e.*, Optison™, Mallinckrodt, Hazelwood, MO; Definity™, DuPont, Pharmaceuticals, Wilmington, DE; EchoGen™, SONUS Pharmaceuticals Inc., Bothell, WA), and these gases have significantly extended the lifetime of these agents (Yasu et al 1997, 1999).

The second potential application for ADV concerns improvements in the generation of both diagnostic and therapeutic US fields. US is well established as a diagnostic imaging modality. However, it suffers from image distortion (loss of contrast and resolution) when tissue is visualized through paths containing spatially varying speed of sound (Freiburger et al. 1992). Changes in the speed of sound cause errors on transmission (focusing) as well as on reception (reconstruction). Phase- and other beam-aberration corrections have been investigated by various groups (Karaman et al. 1993; Ng et al. 1997; Walker and Trahey 1996). Possible solutions include imaging of known and naturally existing structures (*e.g.*, speckle, Ng et al. 1997) and point targets in or near the volume of interest (VOI) (Walker and Trahey 1996; O'Donnell and Flax 1988). The backscatter of those structures is used to adjust the beam forming for that specific site, according to some model of the target structure and the propagation medium. We propose the possible use of vaporized droplets as point targets for phase-aberration corrections. Similar to the proposed approach for the therapeutic application, droplets would be injected IV and acoustically vaporized upstream of the VOI. In low number density, the resulting bubbles would act as point beacons. The reason for the low concentration is to provide single scatterers to avoid multiple

scattering¹ and limit the potential for bioeffects. This application and that of occlusive therapy might not be mutually exclusive. In this case, corrected real-time US imaging should be possible, along with better focusing of ultrasonic fields for therapeutic applications. It is also important that a minimal, but effective, gas bubble diameter be used to avoid infarction, especially when imaging the brain. A minimum gas volume reliable to produce apparently long-term effects, lasting at least 180 min, has been observed in the rabbit brain for carotid bolus injections of 25 μL of air (Helps et al. 1990). For the point target application, we would expect that an injection of 10,000 droplets to the target organ to be sufficient, resulting in a 0.5 μL gas volume (assuming an ADV rate of 100%). The volume is calculated from the normalized droplet size distribution and the maximum size to which the corresponding bubbles would grow.

In addition, the off-resonance scattering based on Medwin (1977) from a 36- μm diameter bubble would provide an estimated 10 dB of scattered signal above brain tissue to act as a point beacon. There is a trade-off in selecting the bubble sizes used. On the one hand, large bubble sizes are necessary to give good backscatter relative to the tissue background and also to last long enough to be imageable. On the other hand, bubbles need to dissolve fast enough to prevent permanent damage to the target tissue, especially in the case of the brain, where there is significant interest in correcting the aberration by the skull. The size of bubbles resulting from the vaporization of perfluorocarbon droplets, including the diffusion of gas from the surrounding liquid, will be examined in this work.

We hypothesize that droplets made by a perfluorocarbon-albumin solution can be stabilized against spontaneous evaporation at body temperature. Furthermore, the resulting agent is also stable to ultrasonic scanning, using a low mechanical index (MI), up to an acoustic pressure threshold above which the agent can be vaporized deliberately.

MATERIALS AND METHODS

Production of droplets

A process for generating droplets was developed, based on mixing saline containing bovine serum albumin and dodecafluoropentane (DDFP, C₅F₁₂, CAS #678-26-2). DDFP has a natural boiling point of 29°C at atmospheric pressure (Simons and Dunlap 1950), which suggests a spontaneous transition into gas bubbles at body temperature. The DDFP droplets were expected to be

¹Multiple scattering makes the correction impossible, since the relative position of the scatterers is unknown. The relative positions highly affect the phase of the incoming signal.

superheated at body temperature and to vaporize when perturbed by an ultrasonic field. Our goal was to produce 36 μm diameter gas bubbles from droplets, based on considerations described above concerning aberration correction for imaging in the brain. A first approximation of the amount of expansion that might occur upon vaporization of a DDFP droplet can be obtained using the ideal gas law $pV = NRT$ (Halliday *et al.* 1997), where p is atmospheric pressure [N m^{-2}], V is volume to be calculated [m^3], N is the amount of droplet material [mole], R is the gas constant $8.31 [\text{J mole}^{-1} \text{K}^{-1}]$, and T is the ambient temperature [K], with the result being a volume expansion factor of approximately 125 or a radial expansion of approximately 5. Consequently, droplets with a diameter of 6–7 μm and less were planned, a size that happens to correspond to the likely maximum for capillary passage (Best and Taylor 1998).

Previous studies have indicated the formation of micrometer-sized droplets of hydrocarbons when solutions with albumin and saline have been sonicated (Suslick and Grinstaff 1990). The approach presented here is similar, but uses an amalgamator, a high-speed vial shaker (MSD Wig-L-Bug, Crescent Dental Mfg. Co. Elgin, IL). Such devices have been used to form perfluorocarbon gas bubbles for US contrast agents (Unger 1995; Unger *et al.* 1996). A series of formulations was tested containing between 1 and 16 mg bovine serum albumin per mL saline. Volumes of a saline-albumin solution (900 μL) and DDPF (100 μL) were injected into an approximately 3-mL auto sampler glass vial. The vial was sealed with a rubber stopper and an aluminium crimp top and then shaken for 30 s using the high-speed vial shaker.

Determination of droplet size

The droplets produced for the different formulations were evaluated using optical microscopy. Samples from each vial were diluted with saline by 1:10 or 1:100, based on the concentration of droplets for the specific formulation, and a portion of this solution was placed on a hemocytometer slide and examined under an optical microscope (Leica Microsystems Inc., Deerfield, IL) at a 400 $\times$ magnification. Optical images were recorded digitally with a CCD camera (Spot, Diagnostic Instruments Inc., Sterling Heights, MI) with an approximate camera resolution of 4 pixels per μm , based on calibration measurements using a stage micrometer (P/N 41416-123, VWR Scientific Products, West Chester, PA). This is 4 to 6 times better than the optical resolution. Droplet sizes were measured within a specified portion of the optical field defined on the hemocytometer grid. Semiautomatic edge and shape detection identified the droplet boundary to determine the perimeter and then to calculate the corresponding droplet diameter. The number density

(i.e., the concentration) was estimated from the volume of fluid from which the droplets were counted by noting the area of the hemocytometer counted and knowing the height of the fluid column in the hemocytometer (100 μm). All individual diameter measures were used to compute individual droplet volumes. Based on the optically-scanned volume (V_{total}), a dilution of 1 to X , and the total volume of droplets counted (V_{droplet}), the droplet volume fraction was then:

$$v = X \cdot \frac{V_{\text{droplet}}}{V_{\text{total}}} \quad (1)$$

Optical observations during droplet heating

Samples of droplets were observed optically during heating by adding 10 μL of a given droplet solution to 10 mL saline in a Petri dish-type container. The dish was then placed in a holder within a water bath (Exacal model, Neslab Instruments Inc., Portsmouth, NH). The holder was positioned below an optical microscope with 320 $\times$ magnification (Nikon SMZ-U, Melville, NY). The droplets lay at the bottom of the dish due to the elevated density of DDFP (1.66 g mL^{-1}) compared to saline. The bath temperature was then raised gradually (4°C min^{-1}) to a maximum temperature of 60°C , while maintaining continuous observation and digital recording through the CCD camera of the microscope for later review.

Vaporization of droplets in vitro

To study the vaporization of droplets in acoustic fields, a water tank with dimensions $18 \times 30 \times 23 \text{ cm}$ was used that contained two flow tube connections mounted on each of the long sides, 9 cm from the far wall and 7.5 cm from the bottom (Fig. 1). Dialysis tubing² (Spectrum Laboratories Inc., Laguna Hills, CA, Spectra/Por membranes, MWCO 12,000–14,000 Daltons) was used as the flow tube. The tank was filled with deionized water and degassed using a multiple pinhole degassing system (Kaiser *et al.* 1996). An external water bath³ (Exacal model, Neslab Instruments Inc., Portsmouth, NH) was used to heat the tank water as well as the circulated fluids to 37°C . The bath's external circulation was connected to a heat exchanger in the water tank to maintain its temperature. The surface of the water in the tank was covered with air-filled plastic balls (P-06821-22, Cole-Parmer Inc., Vernon Hills, IL) to minimize regassing and also to reduce heat loss.

An Erlenmeyer flask (1000 mL) with a saline solu-

²The acoustic attenuation of dialysis tubing was measured following the setup and measurement guidelines in Madsen *et al.* (1986), as less than 1 dB for the range of 1.5 to 8 MHz.

³The word "bath" is used for the external heating system and the word "tank" for the setup that holds the flow tube.

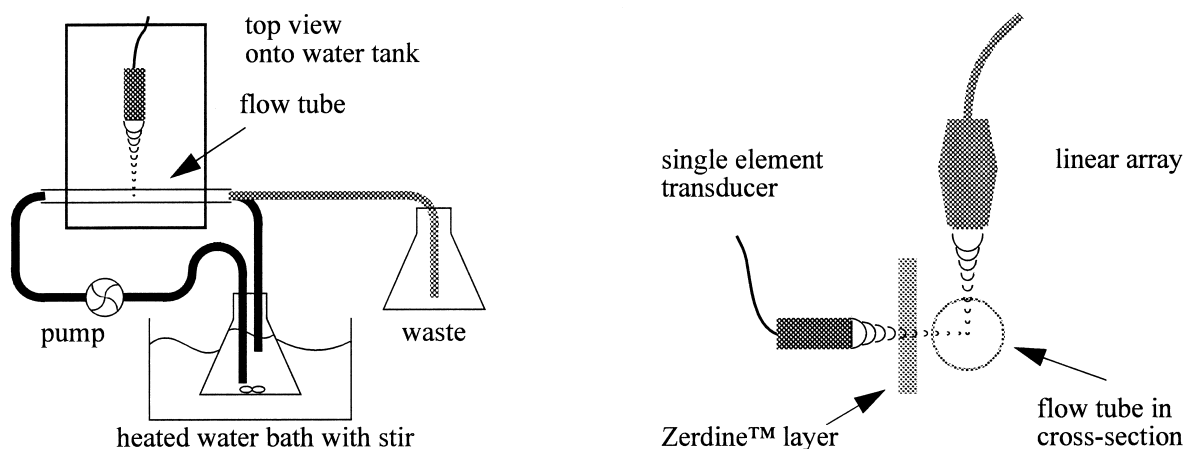


Fig. 1. The setup consists of a water tank (top, on the left) that hosts a flow tube (dialysis tubing) shown in longitudinal and cross-sectional view on the left and right, respectively (not to scale). The single-element transducer is completely submerged. The flow past the outlet on the tank can be directed to a waste container (for one-way flow) or back to the Erlenmeyer flask (for recirculation). All fluids are heated to 37°C.

tion containing droplets was submerged in the water bath and stirred using a magnetic stirrer. The solution of droplets was prepared by adding 100 μL of droplet solution from a shaken vial to 1000 mL of saline. A peristaltic pump (MasterFlex type, Cole-Parmer, Chicago, IL) was used to pump the droplet solution from the flask through the flow tube. Volumetric flow rates from 0.5 up to 2.5 mL s^{-1} could be achieved. For a flow tube of 6-mm diameter, the resulting average velocity was up to 9 cm s^{-1} , assuming a laminar flow condition ($\text{Re} = \rho \cdot d \cdot U / \mu = 600$, where ρ and μ are density and viscosity of the host liquid, respectively, d is the diameter of flow tube, and U is the flow speed) (Kundu 1990). The outlet of the flow tube was connected to a waste container, which was placed at a height of 15 cm above the flow tube. The purpose of this arrangement was to create a slight positive pressure in the flow tube to maintain its inflation.

A linear ultrasonic array (nominally 10 MHz) connected to a diagnostic scanner (Diasonics VST series, Milpitas, CA) was placed above the flow tube, facing downward and aligned with the flow tube (Fig. 1). A single-element transducer (V320, Panametrics Inc., Waltham, MA) with a 14-mm aperture, focused at 76.2 mm) was used to vaporize droplets.⁴ It was mounted at a 90° angle to the imaging plane of the linear array. Both this ADV transducer and the imaging array were focused on the center of the tube. The ADV transducer was controlled by a function generator (model HP-3314, Hewlett-Packard, Palo Alto, CA) and a power amplifier (model 240L or A300, ENI, Rochester, NY). Tone bursts

of typically 3 μs duration with pulse-repetition frequencies from 0.5 to 4.5 kHz were used. The duty cycle was kept below 1%. The ADV transducer was calibrated for its transmitted acoustic pressure using a membrane hydrophone (Reference Shock Wave Hydrophone™, Sonic Industries, Hatboro, PA). The calibration procedure included the acquisition of axial and cross-sectional beam profiles and, for each frequency used, a range of acoustic pressures was measured at the focus.

Observation of gas bubbles

Gas bubbles were monitored as they formed from droplets using ADV. An optical setup, as given in the Materials and Methods section "Optical observations during droplet heating" was used to record images of gas bubbles as they rose in a static fluid (Fig. 8). Based on a video pixel size of approximately 24 μm and a frame rate of 5.6 Hz, rise-time was measured as the distance by which a given bubble moved over the period of N frames. Assuming terminal velocity, the resulting rise speed was used to compute the gas bubble's size (Epstein and Plesset 1950).

Vaporization of droplets in simulated *in vivo* environments

Two key features were added to the experimental flow setup to simulate *in vivo* conditions. Heparinized canine whole blood (6 mL heparin per 250 liter blood) was introduced as circulating fluid and a 2 cm thick layer of attenuating material, Zerdine™ (CIRS Inc., Norfolk, VA), was placed in the beam of the ADV transducer, proximal to the flow tube (Fig. 1). A solution of 75 mL

⁴Here also referred to as ADV transducer.

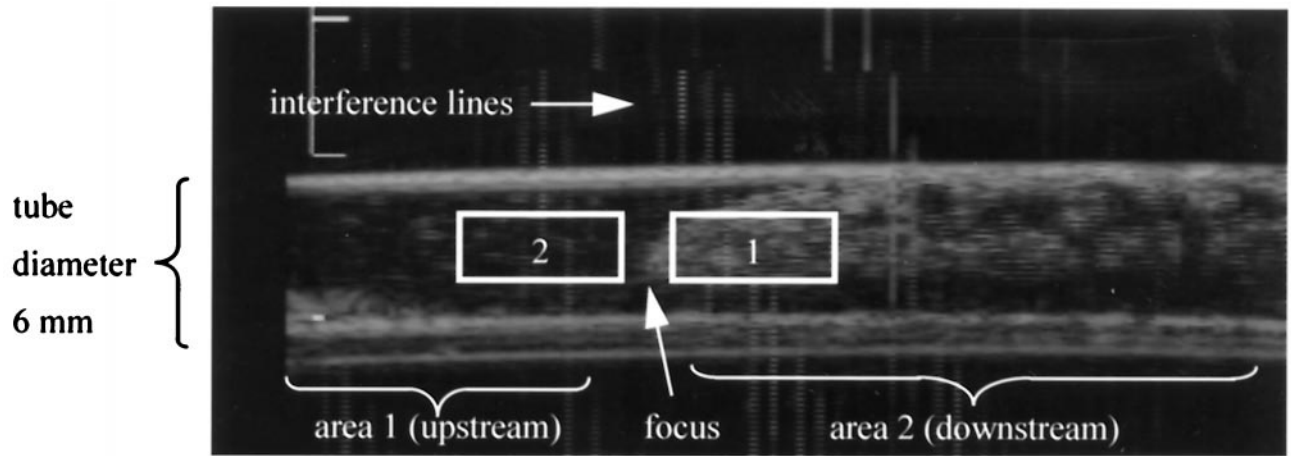


Fig. 2. Ultrasound images taken during ADV, where two areas are selected for the quantification of the echogenicity of the circulated fluid. The mean echo amplitude (MEA) upstream from the focus (area 2) is subtracted from the MEA downstream from the focus (area 1). The resulting value is used as a measure for vaporization because it corrects for echoes from pre-existing scatterers prior to the focus. Acoustical interference is seen as vertical lines in the image. The subtraction method compensates for that artefact. The resulting value is called the relative echo amplitude.

heparinized blood and 50 μL of droplet solution was placed in a 120-mL flask for circulation. The four times higher droplet concentration was used to compensate for the increase in background signal from blood (Fig. 10). Furthermore, the solution was recirculated because the limited amount of blood did not allow a single pass for the constraints of volumetric flow rate and the data acquisition conditions used (see below).

Data acquisition and analysis

Figure 2 shows the dialysis tube in longitudinal view, as imaged by the linear array. The horizontal bright lines are the reflections from the tube's upper and lower walls. The flow direction is from left to right at an average velocity of 9 cm s^{-1} . The droplets were vaporized while passing through the focal zone of the single element transducer located between areas 1 and 2. **The increase in echogenicity was due to the change in the impedance difference between the droplet ($\sim 0.3 \text{ MRayl}$) and the vaporized droplet ($\ll 0.3 \text{ MRayl}$) compared with that of blood.** In area 1, there was a measurable increase in echogenicity, relative to area 2, depending on the amount and size of gas bubbles produced. Interference between the ADV transducer and imaging array appeared as vertical lines (dashed) in the image. Therefore, in the absence of vaporization, there were two sources of signals, a finite echogenicity of the host fluid (due to preexisting gas bubbles and/or acoustically visible droplets) prior to the focus of the ADV transducer, and acoustic interference between ADV transducer and imaging array.

Images, such as the one shown in Fig. 2, were acquired by digitizing the video output of the US machine. Subsequent image analysis, including decompression of the grey scale to linearized echo amplitude, was performed using a MatlabTM script (MathWorks Inc., Natick, MA). The images were acquired at a rate of 2 Hz to allow refill of the evaluation areas 1 and 2 with new droplets/gas bubbles.

$$\text{MEA}(n) = \frac{1}{NM} \sum_{j=1}^M \sum_{i=1}^N A(i, j)$$

$$\text{REA} = \text{MEA}(1) - \text{MEA}(2). \quad (2)$$

where $A(i, j)$ is the signal at pixel $(i, j) \in$ area number n .

The mean echo amplitude (MEA), see eqn (2), of area 2 was evaluated and then subtracted from the MEA of area 1. On average, the resulting value, called relative echo amplitude (REA), was zero for the case where no droplets vaporize. A finite valued REA should occur when droplets evaporate and the echogenicity increases. The time gain control (TGC) was identical for areas at the same imaging depth (*i.e.*, identical for areas 1 and 2). A total of 20 images were taken over 10 s for each sound pressure setting, and the resulting REAs averaged and plotted along with the standard deviation as a function of peak negative pressure.

Thresholds for vaporization from REA as a function of applied pressure were determined by least square fitting to the following two line segments.

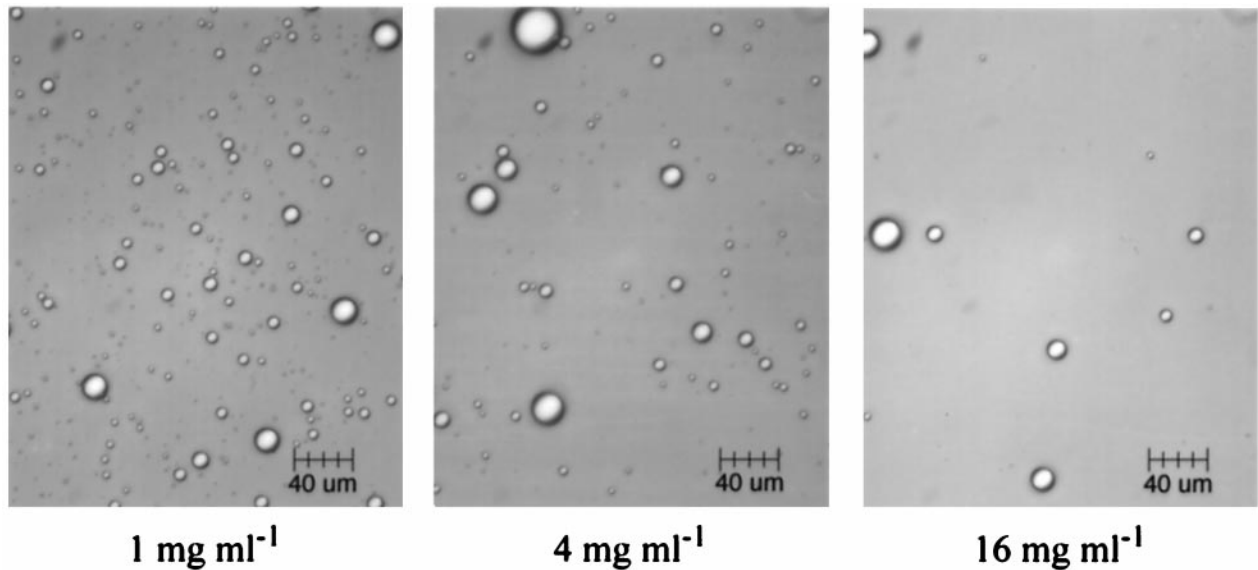


Fig. 3. The three microscope pictures show droplet formation for 1, 4, and 16 mg albumin per mL saline. The concentration of DDFP was the same in all three cases (0.1 mL mL^{-1}). A lower concentration of albumin results in smaller droplets, as well as in a larger number density. The pictures have been taken for the same dilution, as well as with the same magnification. Note that, qualitatively, the average droplet size changes with albumin concentration. The picture on the left shows many small droplets and a few larger droplets. The picture in the middle has fewer small droplets, and the picture on the right has no apparent small droplets. The size of the largest droplets, however, does not change significantly. Note that, for a constant amount of fractional droplet volume ($\sim R^3$), the cross-sectional area ($\sim R^2$) of droplets increases with decreasing mean droplet size. Therefore, one sees more “white” in the image with smaller mean droplet size.

$$\begin{aligned} \text{REA}(p) = & (a_1(p - p_{\text{threshold}}) + b) \cdot w(p, 0, p_{\text{threshold}}) + \dots \\ & (a_2(p - p_{\text{threshold}}) + b) \cdot w(p, p_{\text{threshold}}, p_{\text{max}}) \end{aligned} \quad (3)$$

$$w(p, p_1, p_2) = \begin{cases} 1 & p \in [p_1, p_2] \\ 0 & \text{otherwise} \end{cases}$$

The structure of the fitting function was fairly simple. A line with slope a , intercept b and shifted origin was multiplied by a window function that limited the extension of the line to all the data points before/after the pressure threshold to be found.

RESULTS AND DISCUSSION

Generation of droplets

When saline and DDFP were shaken without albumin, the resulting mixture initially contained droplets, but the droplets coalesced substantially within a few seconds; the mixture appeared to return rapidly to its original state of two separate bulk liquids. In contrast, mechanically-shaken mixtures containing albumin formed droplets that, at least in part, remained as an emulsion for several weeks, based on continued observation of ADV using samples stored under refrigeration at 4°C . Figure 3 shows droplets made for three different

concentrations of albumin (16, 4, and 1 mg mL^{-1} saline). Using the optical methods above, the size distribution for droplets formed with 1 mg albumin per mL saline is

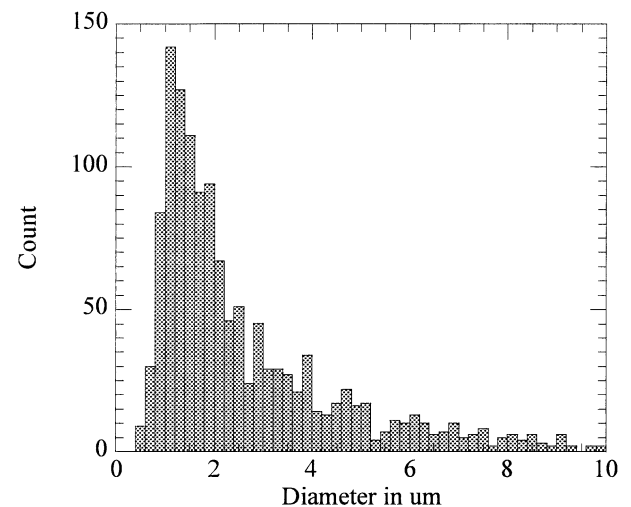


Fig. 4. Size distribution for DDFP droplets made with 1 mg albumin per mL. The mean size is $1.5 \mu\text{m}$ radius. The total number of droplets was 1332 for an approximately $3 \times 10^{-5} \text{ mL}$ sample volume, corresponding to a concentration of 4.82×10^7 droplets per mL.

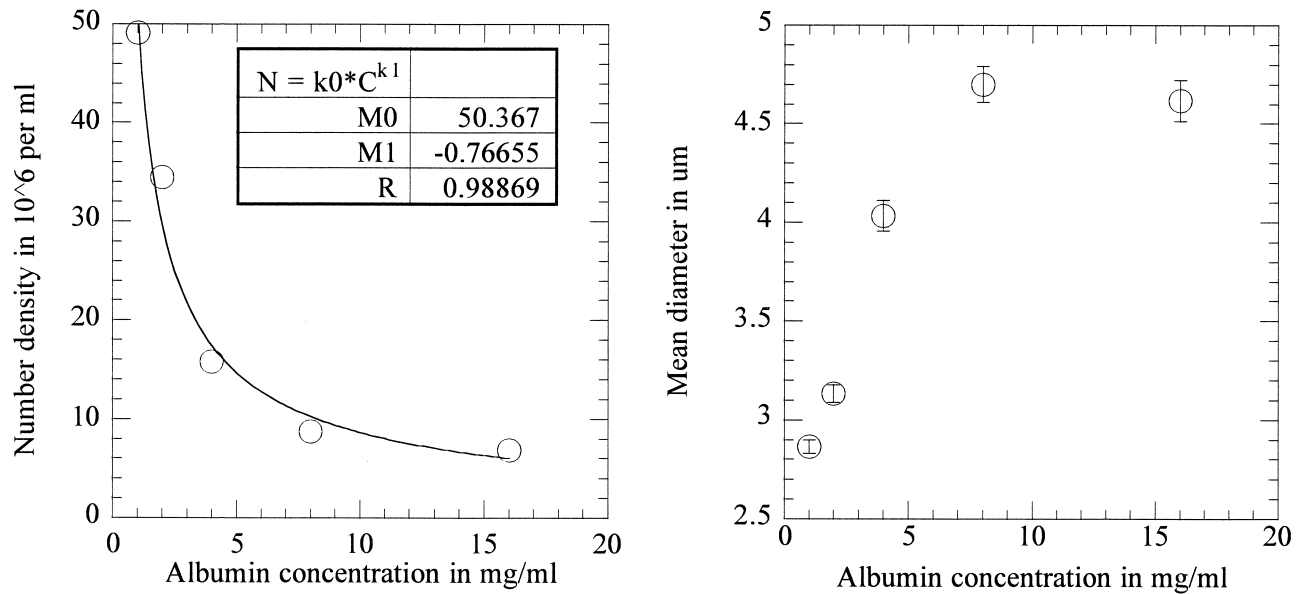


Fig. 5. The number density (left side) and the mean radius (right side) of droplets are plotted as a function of the albumin concentration. The number density decreases, and the mean radius increases, with increasing albumin concentration. The average volume of DDFP incorporated into droplets remained essentially constant.

given in Fig. 4. Note that this emulsion should be substantially transpulmonary based on its size (*i.e.*, it should pass the capillary bed of the lungs) because 90% of the droplets are smaller than $6 \mu\text{m}$ in diameter. Figure 5 shows that the number density decreases with increasing albumin concentration. Based on the general trend in the data, the points were fit by least squares methods to a power law, $N = k_0 C^{k_1}$, where N is the number density, C is the concentration of albumin and k_0 and k_1 are constants. The coefficients were found to be $k_0 = 50.4$ and $k_1 = -0.767$ ($R^2 = 0.978$). The mean droplet size increased by $\sim 62\%$ from $2.9 \mu\text{m}$ mean diameter to $4.7 \mu\text{m}$ over the range of albumin concentrations used. The decrease in number density and the increase in size are mutually dependent, because the percentage of DDFP incorporated into droplets remained nearly constant at approximately 30%. Note that, if the total volume fraction of droplets remains the same, the total surface area formed by droplets increases when the average droplet size decreases. In that case, the amount of albumin used to stabilize the surface of the droplets is greater. But, the experiment shows that, when a lower concentration of albumin is used, smaller droplets form. This might seem paradoxical, but could be explained if the albumin used is at a saturation level for droplet stabilization and changes in interfacial surface tension and viscosity of the liquid influenced the size distribution. Suttiprasit *et al.* (1992) showed that the surface tension of bovine serum albumin solutions increases with increasing serum concentration, and Ni *et al.* (1998) measured a 35% increase

in droplet size for methylmethacrylate suspensions when the interfacial tension was doubled. Our results are consistent with these trends.

Droplet vaporization and frequency dependence

Under optical observation, droplets in static solution did not vaporize when heated to a maximum of 60°C (limit of the heating bath). However, as indicated in Fig. 2, droplets did vaporize in the presence of an ultrasonic field, producing increased echogenicity. So the immediate question was: "Why do they boil in the ultrasonic beam?" A possible mechanism could be inertial acoustic cavitation, whose frequency-dependence for onset has been predicted according to the model of the mechanical index (MI) given by Apfel and Holland (1991):

$$MI = \frac{\text{peak negative pressure in MPa}}{\sqrt{\text{frequency in MHz}}}. \quad (4)$$

If cavitation was the mechanism for droplet evaporation, then the MI at threshold should be a constant and the peak negative pressure needed would be larger for higher frequencies. Accordingly, the threshold for ADV, in peak negative pressure, was measured as a function of frequency in a series of measurements performed using single-element transducers to transmit short tone bursts at a range of center frequencies, as described in the Methods section.

For each frequency, the relative echo amplitude was

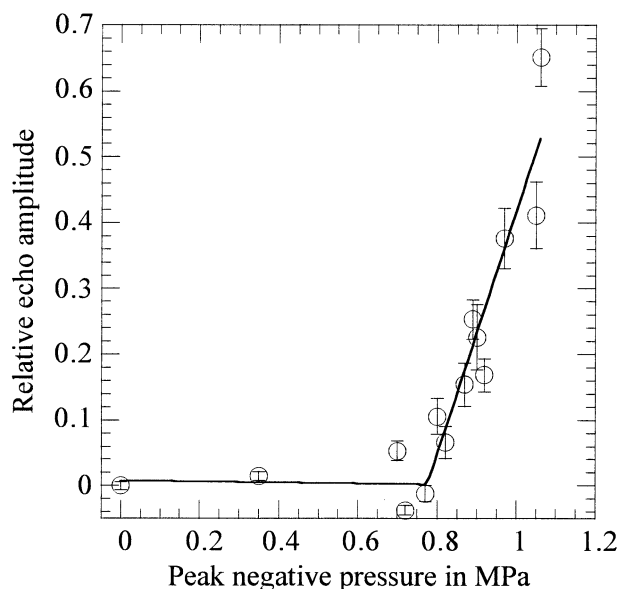


Fig. 6. The plot shows the relative echo amplitude as a function of applied peak negative pressure. The frequency used was 7.6 MHz.

measured as a function of the peak negative acoustic pressure (see Methods section, as well as Fig. 2). Figure 6 shows results for a focused transducer driven by a 7.6-MHz, 5-cycle tone burst, with a 0.2-ms pulse-repetition period (PRP). Each data point in the chart is an average for 20 measurements of the increased echo amplitude due to ADV. The relative echo amplitude remains at zero with increasing applied acoustic pressure, until the pressure approximately equals 0.7 MPa. At that point, the curve shows a much greater slope. The pressure threshold of 0.7 MPa at 7.6 MHz corresponds to a mechanical index of 0.25, which is well below the maximum available with commercial US systems.

A comment on the functional dependence of the relative echo amplitude as a function of pressure should be given. It has been observed that the relative echo amplitude slightly increased while the applied pressure was below "threshold for ADV." If the subtraction process, eqn (2), is working, why should this happen? In the case of a 1 mg mL^{-1} droplet emulsion, insonified at 8 MHz, the slope below threshold had been up to 14% of the slope after the threshold. We believe that this might be because of the distribution of droplet sizes and the high frequency provides additional scattering from the ADV transducer into the imager.

In a similar manner, the threshold for vaporization was measured at several frequencies ranging from 1.5 to 8.0 MHz. These results are shown in Fig. 7, in which it should be noted that the threshold is not a constant MI

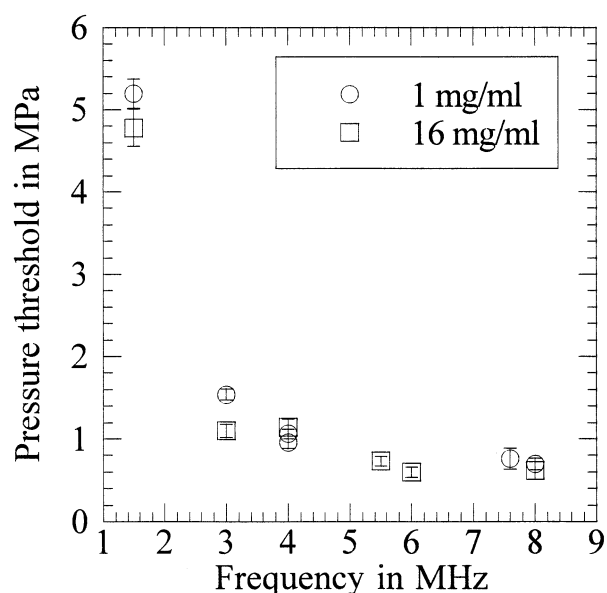


Fig. 7. The frequency-dependence of ADV. For each frequency, REA was fit by means of least square to two line segments; see Fig. 6, for example. The pressure threshold for ADV was found from the pressure amplitude where the two segments intersect.

value but, rather, even lower at higher frequencies. Even though some of the acoustic pulse parameters, like PRP, duty cycle and beam geometry, were not the same for the different measurements taken, the threshold appears to change strongly and consistently with frequency; if accurate, this dependence is not modeled well by the $f^{-1/2}$ frequency dependence.

Observation of gas bubbles

Gas bubbles made by ADV (7.6-MHz, 5-cycle tone burst, 0.9-MPa peak negative pressure, 0.2-ms PRP) were observed optically as described in the Methods section. The droplets used were from the 1 mg mL^{-1} albumin formulation diluted into saline (1:10,000). Figure 8 shows the overlay of 15 optical images from the microscope, when focused inside the tube where droplets were acoustically vaporized.

Several bubble traces are visible, such as the traces of bubble 1 and bubble 2. Their diameters were measured to be $\sim 108 \mu\text{m}$ and $\sim 91 \mu\text{m}$, respectively based on rise time estimates. Several other bubbles were measured, ranging from 75 to $150 \mu\text{m}$ diameter. Based on the droplet size distribution, such large bubbles could not be explained with ordinary droplet-to-gas bubble conversion using a volume expansion factor of 125 based on an ideal gas law calculation. In addition, a simple measurement of the volume expansion of evaporating pure DDFP

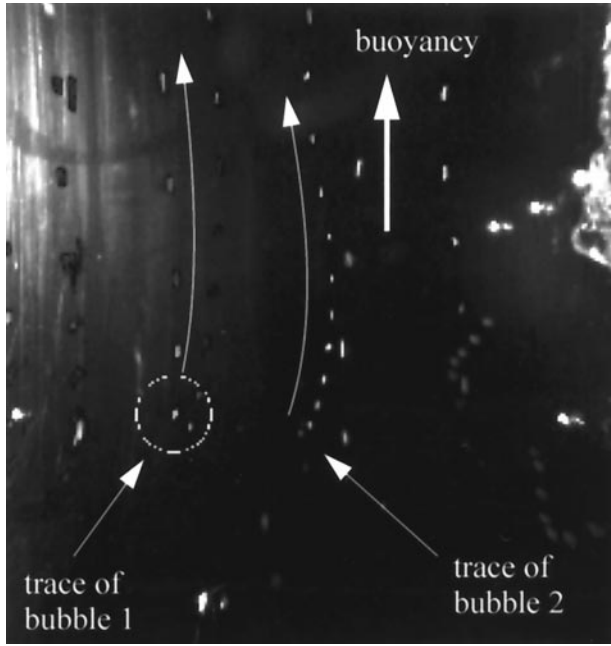


Fig. 8. Gas bubbles formed from evaporated droplets have been monitored under a microscope. The picture shows multiple, superimposed frames of a movie. Based on the microscope's magnification, bubble 1 is approximately $100\ \mu\text{m}$ in diameter. Assuming simple drag force, the frame rate used (180 ms per frame) yields a bubble size of $108\ \mu\text{m}$; whereas bubble 2 yields $91\ \mu\text{m}$.

was found to be approximately 151 at 37°C . Such a volume expansion factor would correspond to an increase in diameter by only a factor of 5.3 (i.e., a droplet as large as $10\ \mu\text{m}$ diameter expanding to only a $53\ \mu\text{m}$ diameter bubble of pure DDFP). A reason for further expansion is intake of dissolved gases from the surrounding fluid. An initial gas bubble contains only DDFP gas. The host liquid is air-saturated, containing dissolved N_2 , O_2 , CO_2 , and Ar. A simulation using the formulation of Neppiras (1980), eqn (5), was extended to multiple gases, eqn (6), to predict the uptake of gas into the bubble:

$$\dot{R} = \frac{C_0 D}{\rho} \left(\frac{C_\infty}{C_0} - 1 - \frac{2\sigma}{p_0 R} \right) \left(\frac{1}{R} + \frac{1}{\sqrt{\pi D t}} \right), \quad (5)$$

where R_0 is the current gas bubble radius, C_0 and C_∞ are saturation concentration and concentration at infinity in the surrounding host liquid, respectively, D is the gas's constant of diffusion, ρ is the density of the gas inside the gas bubble, σ is the surface tension between gas and liquid, p_0 is the ambient static pressure, and t is time. For multiple gases,

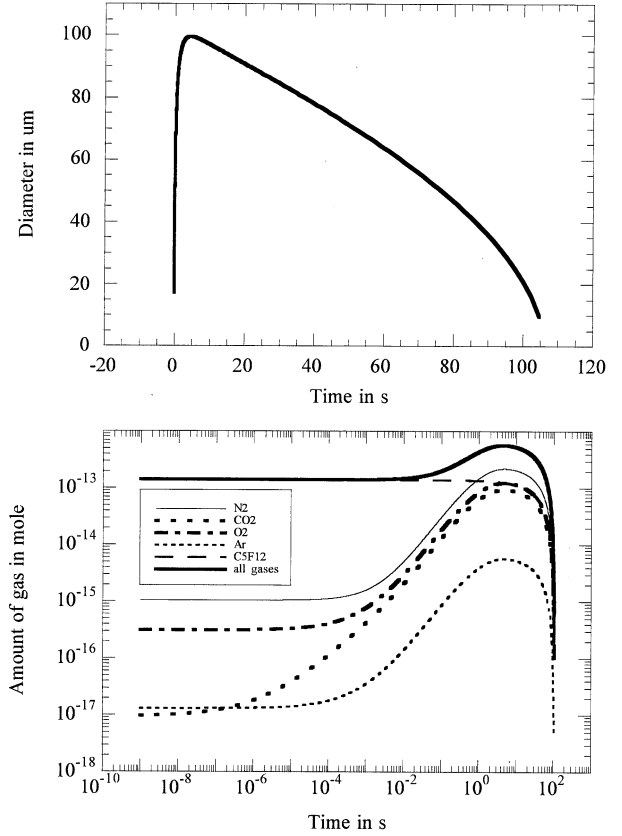


Fig. 9. Top graph shows the diameter of an initially $17\text{-}\mu\text{m}$ diameter DDFP bubble as it expands due to gas diffusion from the host liquid (time interval from 10^{-9} to ~ 5 s), as well as subsequent dissolution of the gas bubble (time interval from ~ 5 s to 100 s), see eqn (6). Bottom graph shows the amount of the individual gases as they diffuse in and out of the bubble.

$$\dot{n}_j(t) = \frac{4\pi R^2(t)}{M_j} D_j \left(C_{j,\infty} - \text{Min} \left(C_{j,0}, \frac{n_j(t)}{(4/3)\pi R^3(t)} \right) \right) \times \left(1 + \frac{2\sigma_j}{p_0 R(t)} \right) \left(\frac{1}{R(t)} + \frac{1}{\sqrt{\pi D_j t}} \right) \quad (6)$$

$$\dot{R}(t) = \frac{1}{4\pi R^2(t)} \frac{1}{N} \sum_{j=1}^N \frac{\dot{n}_j(t) M_j}{\rho_j} = \frac{R(t)}{3} \frac{1}{N} \sum_{j=1}^N \frac{\dot{n}_j(t)}{n_j(t)},$$

where n_j is the current concentration of gas j inside the gas bubble, M_j is the molar mass of gas j , and N is the total number of gases. (A similar formulation for contrast agents can be found in Kabalnov *et al.* 1998a, 1998b.)

Using the experimentally-observed DDFP volume expansion of 151 (diameter expansion of 5.3), a droplet with an initial diameter of approximately $3\ \mu\text{m}$ will form a $17\text{-}\mu\text{m}$ diameter gas bubble of pure DDFP. A $17\text{-}\mu\text{m}$ DDFP bubble expands, due to inflowing gases, to 100

Table 2. Numerical results of the fitting algorithm on ADV data for simulated *in vivo* environments.

Case	Frequency (MHz)	Pressure threshold (MPa for 16 mg mL ⁻¹)	Pressure threshold (MPa for 1 mg mL ⁻¹)	Frequency (MHz)	Pressure threshold (MPa for 16 mg mL ⁻¹)	Pressure threshold (MPa for 1 mg mL ⁻¹)
Droplets in saline	1.5	4.78 (0.23)	5.20 (0.18)	4.0	1.14 (0.10)	1.00 (0.08)
Droplets in saline plus Zerdine™	1.5	5.49 (0.24)	—	4.0	—	1.71 (0.01)
Droplets in blood	1.5	5.42 (0.18)	6.09 (0.07)	4.0	—	1.12 (0.05)
Droplets in blood plus Zerdine™	1.5	6.21 (0.16)	—	4.0	—	—

Standard deviation values are given in parentheses.

μm diameter (see top graph in Fig. 9). This could explain quite well the occurrence of the 91- and 108- μm diameter gas bubbles seen in Fig. 8. The bottom graph of Fig. 9 shows the contributions of the individual gases. The gas bubble is completely dissolved after approximately 100 s. In the case of using bubble formation to occlude blood vessels, the lifetime could possibly increase for several reasons. The formation of bubble clouds collected in the vessels may reduce the gas diffusion due to the exclusion of surrounding blood, the reduction of diffusion boundaries (gas-blood interface) and an increasing local concentration of DDFP. Bubbles may coalesce into larger bubbles when in sufficient numbers for bubble-bubble contact. There is likely a reduced diffusibility and/or convection of gases in tissue compared to pure water. However, the flow of fluid around the bubbles may increase the dissolution of bubbles due to the disruption of the concentration gradient in the liquid surrounding the bubbles. The calculations presented here are for the case of a static fluid, which may only exist after the blood flow is reduced, or for the case when it can be considered that the fluid is essentially stationary with the respect to the bubble (bubble substantially moves with the fluid).

It should be mentioned again that very sparse blockage will most likely not affect the tissue because collateral flow compensates for a reduction in the primary supply vessels. In the case of nuclear perfusion scans, macroaggregated albumin particles (MAA) are used as tracers. These particles (26- μm diameter) are clearly larger than capillaries. However, side effects from particles transit to the arterial system by bypassing the capillaries of the lung via RL shunts (*i.e.*, vessels that bypass the lung by effectively connecting the right ventricle to the left atrium), have not been detected or are, at least, not common (Potchen and Evens 1971; Davis and Taube 1978).

ADV in simulated *in vivo* environments

Where penetration is an issue, lower frequencies are generally better for medical applications because layers of sound-absorbing tissue have to be overcome; a typical

value for the acoustic attenuation in soft tissue is 0.5 dB MHz⁻¹ cm⁻¹. However, in the case of using US for the vaporization of droplets, the pressure threshold for ADV is significantly reduced at higher frequencies, as shown in Fig. 7. To use ADV for *in vivo* applications, one would have to penetrate tissue layers between the US transducer and the target (*i.e.*, the tissue or vessel of interest). The attenuation of a 2-cm tissue path with 0.5 dB MHz⁻¹ cm⁻¹ acoustic attenuation at 1.5 MHz and 4 MHz is 1.5 dB and 4 dB, respectively. However, because the threshold of ADV at 1.5 MHz is more than 4 times higher than at 4.0 MHz, higher frequencies may, nevertheless, be preferred. Experiments to simulate the *in vivo* situation by overlaying 2 cm of Zerdine™ were performed at 1.5 MHz and 4 MHz.⁵ As one would expect, a greater acoustic output was necessary for ADV when Zerdine™ was in place. For droplet emulsions containing 16 mg albumin per mL saline (type A) and tone burst centered at 1.5 MHz, the threshold for ADV increased from approximately 4.78 to 5.49 MPa (*i.e.*, 15%) (see Table 2). For comparison, assuming a linear dependence on peak negative pressure, the theoretical increase due to a 1.5 dB signal amplitude attenuation is 19%. A control measurement for the same range of sound pressures, but in the absence of droplets, was made. The REA remained zero and is interpreted as indicating the absence of detectable bubbles in the host liquid, although some cavitation could be occurring and not be imageable. For ADV in saline, using a droplet emulsion with 1 mg albumin per mL saline (type B) and tone bursts centered at 1.5 MHz, the pressure threshold for ADV was found to be 5.20 MPa (*i.e.*, 9% greater than the type A emulsion with the greater mean droplet size), where the Laplace pressure acting on the droplets is smaller.

At 4 MHz, the pressure threshold for ADV was measured for several cases. For type B droplet emulsion, the threshold increased from 1 MPa peak negative pressure to 1.71 MPa (*i.e.*, a 71% increase, theoretically

⁵The following experiments that simulate *in vivo* environments were performed with recirculation of blood and, for comparison, also recirculation of saline.

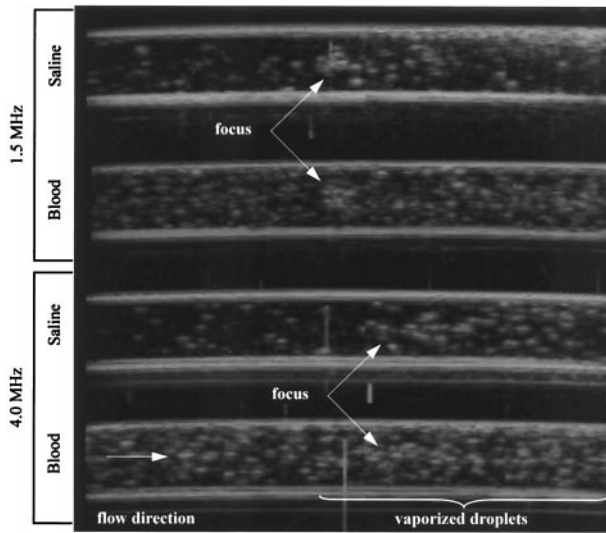


Fig. 10. Grey-scale pictures taken for ADV in saline and blood using tone bursts at 1.5 and 4.0 MHz. One can see the increase in background signal when blood is used as the circulating fluid. However, the vaporization of droplets is still observable, as echogenicity increases above the background signal from blood.

58%), when two layers of Zerdine™ (1 cm each) were mounted in front of the flow tube. This means that we were able to use ADV while penetrating 2 cm “tissue” at 4 MHz. Furthermore, it seems that frequencies of approximately 4 MHz rather than its harmonics at 8 MHz were controlling the vaporization because the increase in

threshold was consistent with the overlying attenuation computed for 4 MHz. The threshold for using type A droplet emulsion increased also, but only to 1.12 MPa (*i.e.*, 14%), which is on the order of the standard deviation (see Table 2).

In another series of experiments, the host liquid was changed from saline to heparinized whole blood (50% hematocrit) to achieve a better simulation of the *in vivo* situation. The REA was again measured as a function of sound pressure. Acquired flow tube pictures are shown in Fig. 10. A significant increase of background signal due to the blood was observed. Measurements were taken of REA to determine the threshold of ADV, using tone bursts at 1.5 and 4 MHz. The pressure dependence of REA was evaluated and is given in Fig. 11. The acoustic pressure threshold for ADV at 1.5 and 4 MHz using a 1 mg mL⁻¹ droplet emulsion was found to be 6.1 MPa and 1.1 MPa, respectively. At 1.5 MHz, the threshold for a 16 mg mL⁻¹ droplet emulsion was measured as 5.4 MPa, 11% less than for the 1 mg mL⁻¹ droplet emulsion. Introducing Zerdine™ into the acoustic beam, the pressure threshold increased from 5.4 MPa to 6.2 MPa (*i.e.*, 15%).

SUMMARY

Generating droplets

It has been shown that stabilized droplets can be made from DDFP. The mean size and number density could be changed by altering the droplet’s formulation. At the present time, the droplet yield is approximately 30%. The resulting agents, as described here, could

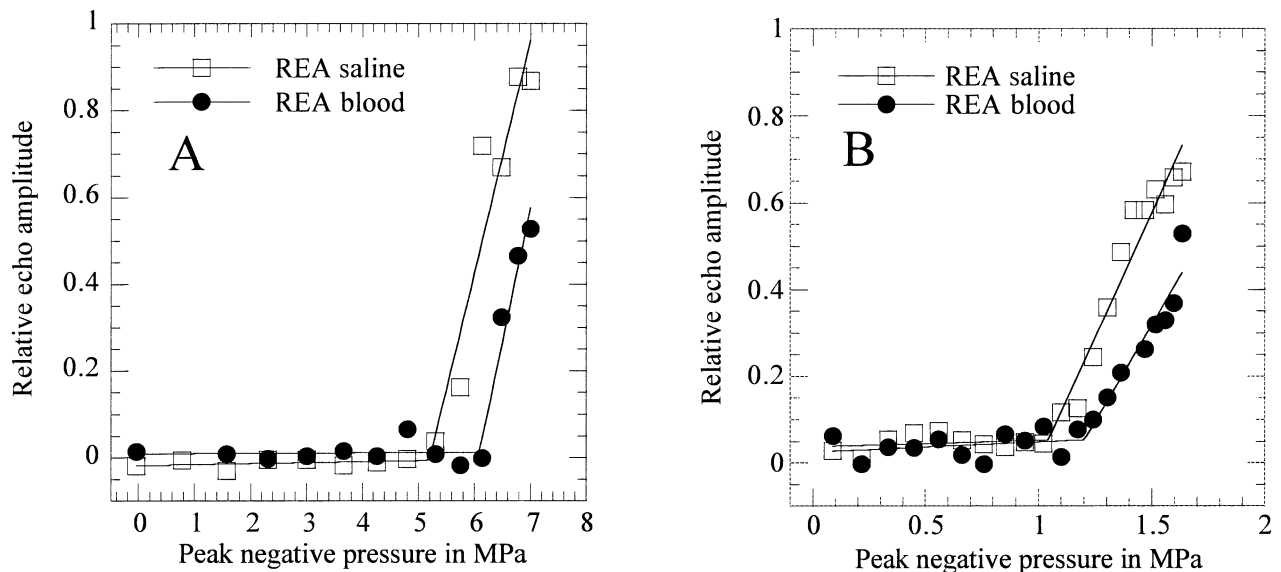


Fig. 11. Acoustic vaporization of droplets in blood using a short tone burst of (A) 1.5 and (B) 4.0 MHz. The relative echo amplitude for ADV was fitted by a least-square piecewise linear function. See Table 2 for thresholds for ADV.

likely be manufactured to be compatible biologically because similar perfluorocarbon liquids are commonly used as blood substitutes (Riess and Krafft 1999) and EchoGen™, an agent of very similar chemical composition, is already in use in clinical trials (Correas and Quay 1996; Quay 1996). Pretreatment and spontaneous evaporation causes EchoGen™ to become an injectable dispersion of microbubbles. Droplets made in our laboratory also show spontaneous conversion, after they are introduced to an environment with a temperature above their boiling point. However, only a small fraction of them boil, leaving stable droplets that are in a superheated state. A pretreatment scheme similar to that of EchoGen™ could eventually be used to limit the possibility of spontaneous vaporization to a nonrisk level at which collateral blood flow can fully compensate for any small-scale temporal occlusion. In addition, human serum albumin (HSA) has also been used to stabilize US contrast agent with perfluorocarbon gas (Optison®, Mallinckrodt Medical, St. Louis, MO). It is expected that droplets of the type described here will also be formed in solutions of HSA, but this has not yet been tested.

Vaporizing droplets

Experiments showed that droplets superheat up to 60°C, well beyond the natural boiling point of pure DDFP of 29°C. Droplets in the superheated state can be vaporized in an acoustic field. Pressure variations in the transmitted acoustic signal showed that an acoustic pressure threshold for ADV exists, and measurements at various frequencies in the range from 1.5 to 8 MHz showed that the threshold apparently decreases with increasing frequency. This leads to a trade-off situation in medical applications, in which overlying tissue attenuation would favor the use of lower frequency US. So far, we have vaporized droplets in simulated *in vivo* environments at 1.5 and 4 MHz. The frequency of choice for real *in vivo* ADV depends on factors such as attenuation by overlying tissues. For the presented case of 0.5 dB MHz⁻¹ cm⁻¹, an ADV transducer operating at 4 MHz would be preferable over one at 1.5 MHz, up to a penetration depth of 10 cm.

It has been observed that the acoustic pressure threshold for ADV decreases with increasing frequency, which is opposite to what is known for the frequency-dependence of acoustic cavitation. If acoustic cavitation is not the mechanism by which the droplets boil, the underlying physical mechanism has yet to be determined. There are phenomena in the droplet literature (*e.g.*, deformation of droplets in acoustic fields) that increase with increasing frequency, which might explain the characteristics as seen in our experiments (Tian et al. 1993).

Further investigations are needed to achieve ADV *in vivo* and to study the extent to which blood vessels are

occluded from resulting gas bubbles. From a medical point of view, safety issues such as size distribution, number density, and lifetime of the bubbles produced, as well as stability of droplets in the absence of US, are very important.

Furthermore, phenomena warranting study include rate of vaporization and gas bubble size distribution, as well as sensitivity to changes in blood pressure, blood constitution and temperature.

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